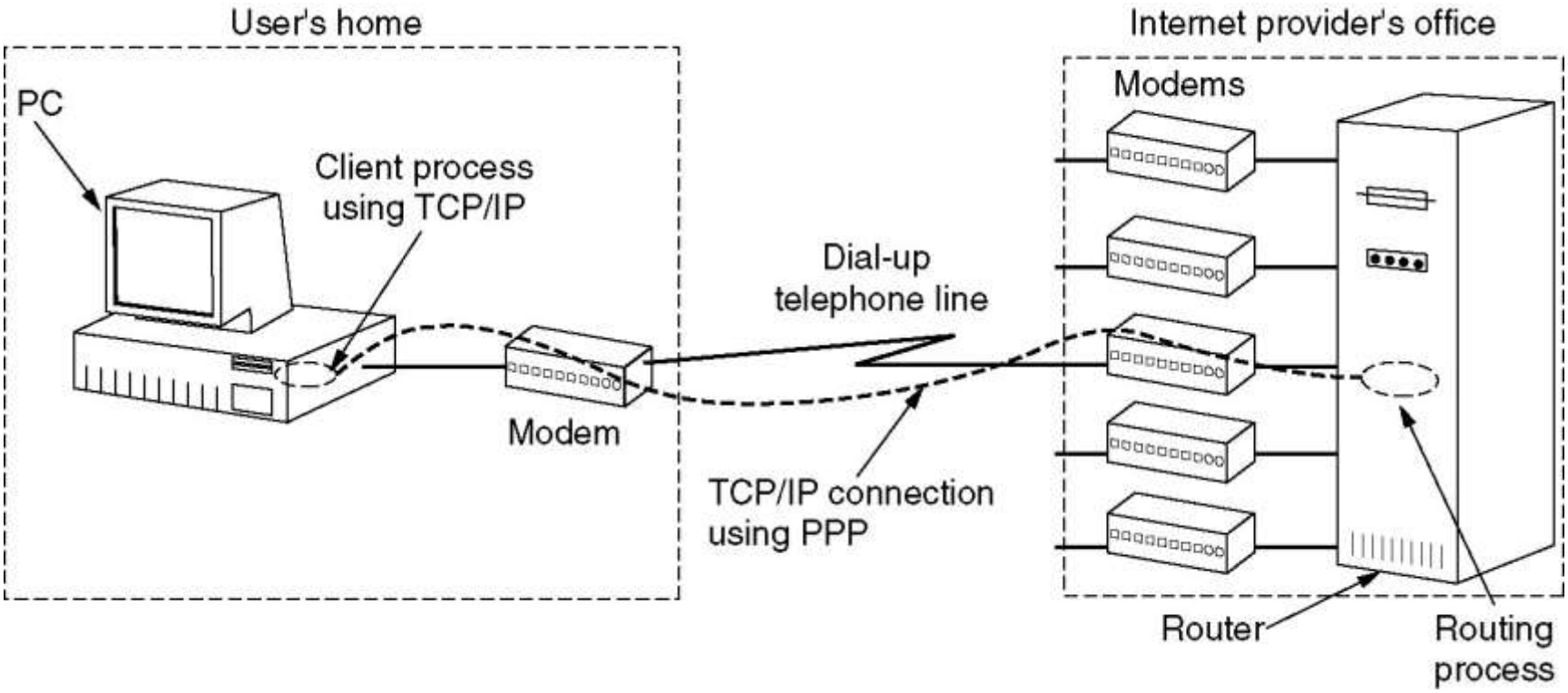
HDLC and PPP

The Data Link Layer in the Internet



Technology like Ethernet cannot provide “high-level” functionality like connection management and parameter negotiation

Point to Point Data Link Control

- One sender, one receiver, one link: easier than broadcast link:
    - No Media Access Control
    - No need for explicit MAC addressing
    - E.g., dialup link, ISDN line
- Popular point-to-point and high-level DLC protocols:
    - PPP (point-to-point protocol)
    - HDLC: High level data link control (Data link used to be considered “high layer” in protocol stack). HDLC is also used in multi-point links (one station many receivers)
- These protocols can often be run over other data link technologies providing best of both worlds
  - E.g., PPPoE, HDLC encapsulation by Ethernet

PPP Design Requirements [RFC 1557]

Functionality : (similar to link layer services + extra management functions)

- **Packet framing -** encapsulation of network-layer datagram in data link frame
- – **Multi-protocol** - carry network layer data of any network layer protocol (not just IP) *at same time* ability to demultiplex upwards
- **Bit transparency -** must carry any bit pattern in the data field (even if underlying channel can't)
- **Error detection** - not correction

PPP Design Requirements (cont.)

The extra stuff:

- Connection liveness: detect, signal link failure to network layer
- Network layer address negotiation: endpoint can learn/configure each other's network address and other characteristics.
- Authentication: who are you (or at least whose account do I bill for your dial-in time?)
  - This information is used by traffic management software to control bandwidth to individual subscribers
- Management features: loopback detection

PPP non-requirements

- No error correction/recovery

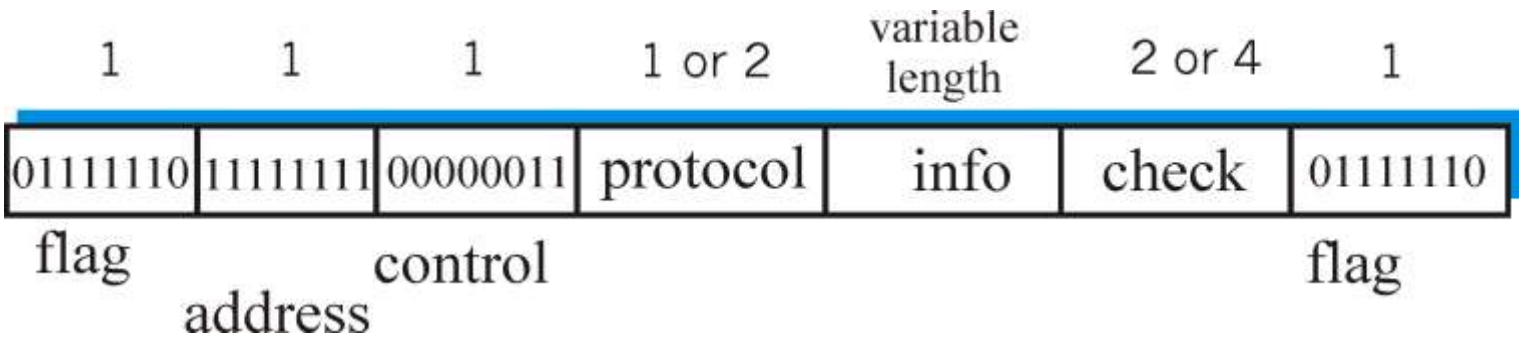
  (modems do one layer FEC, one layer packetization + retransmission “under the covers” anyway; other technologies are pretty reliable)
- No flow control
- Out of order delivery OK

Error recovery, flow control, data re-ordering

all relegated to higher layers!

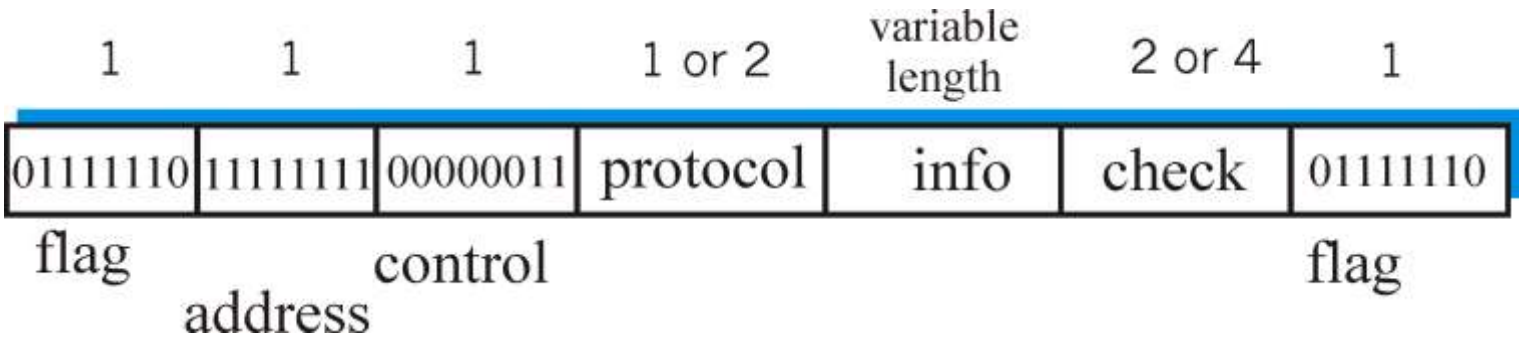
PPP Data Frame

- Flag: delimiter (framing)
- Address: ignored. (historical)
- Control: ignored. (historical)
- Protocol: upper layer protocol to which frame delivered (e.g., PPP-LCP, IP, IPCP, etc)

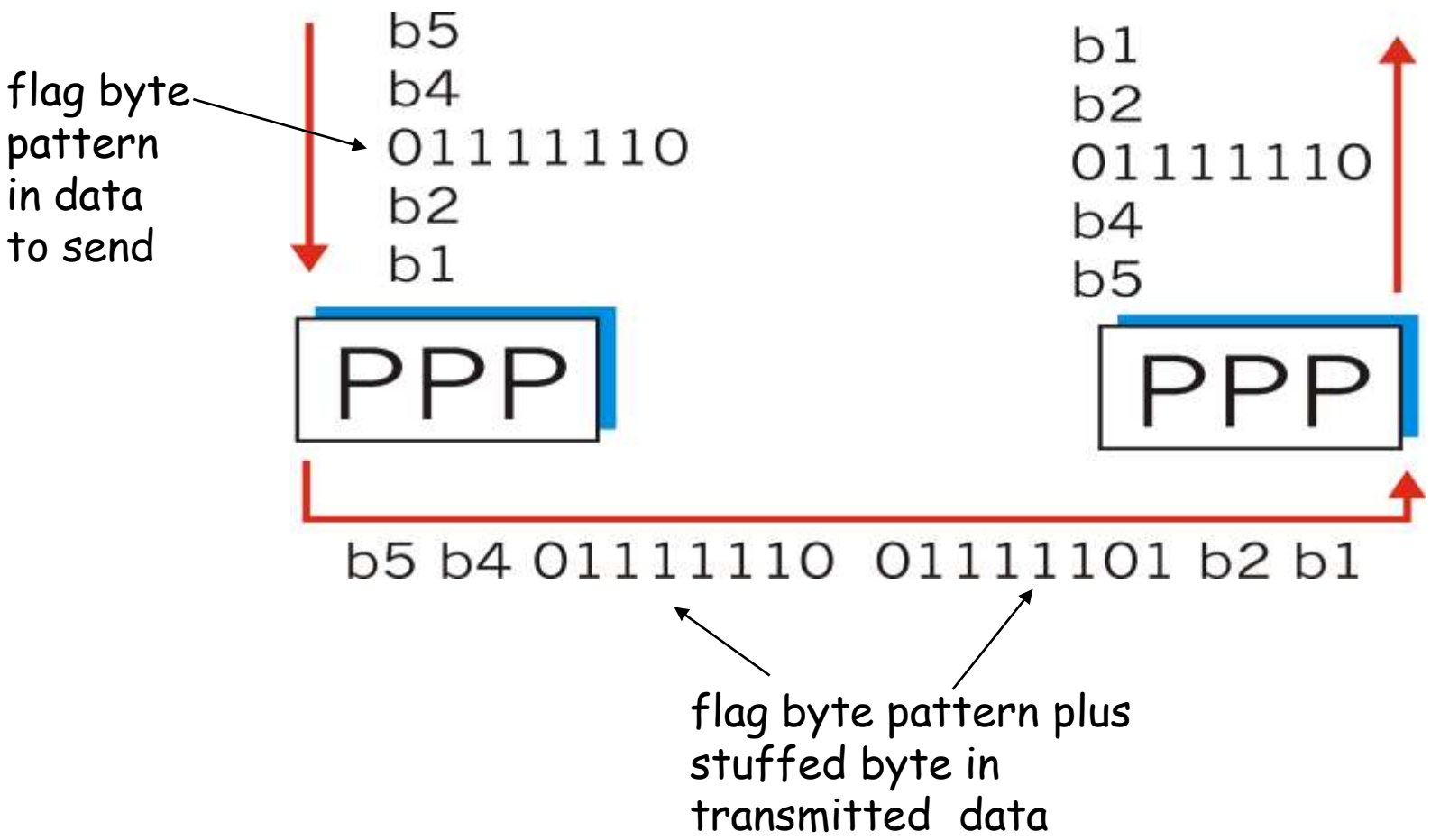


PPP Data Frame

- info: upper layer data being carried
- check: cyclic redundancy check for error detection



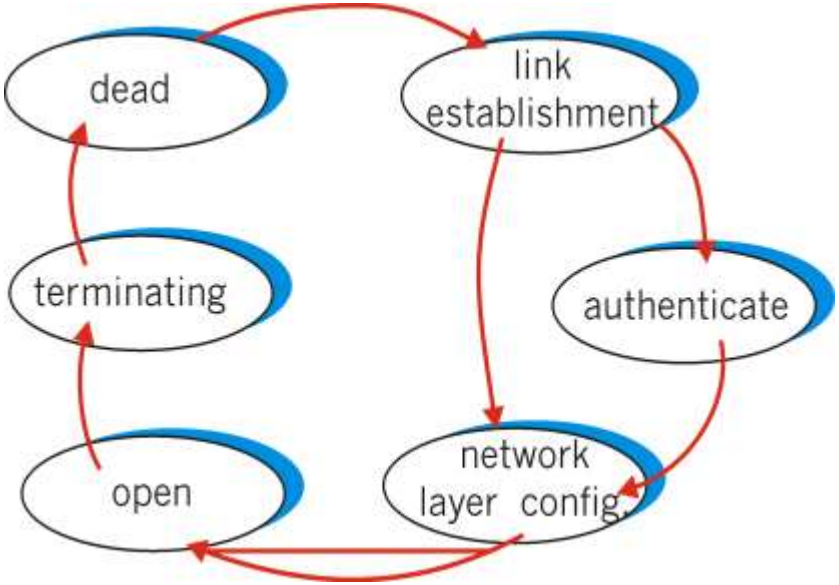
Byte Stuffing



PPP Data Control Protocol

Before exchanging network-layer data, data link peers must

- Configure PPP link (max. frame length, authentication)
- Learn/configure network layer information
  - for IP: carry IP Control Protocol (IPCP) msgs (protocol field: 8021) to configure/learn IP address



Where does PPP get used?

- Dial-up – PPP over async serial, over modem
- ADSL – PPP over Ethernet
- Backbone – Packet over SONET (POS)
- Why?
  - Framing (dialup, POS)
  - Efficiency (POS)
  - Authentication, address negotiation (PPPoE)

High-Level Data Link Control (HDLC)

- HDLC was defined by ISO for use on both point-to-point and multipoint data links.
- It supports full-duplex communication
- Other similar protocols are
  - Synchronous Data Link Control (SDLC) by IBM
  - Advanced Data Communication Control Procedure (ADCCP) by ANSI
  - Link Access Procedure, Balanced (LAP-B) by CCITT, as part of its X.25 packet-switched network standard

HDLC Overview

Broadly HDLC features are as follows:

- Reliable protocol
  - selective repeat or go-back-N
- Full-duplex communication
  - receive and transmit at the same time
- Bit-oriented protocol
  - use bits to stuff flags occurring in data
- Flow control
  - adjust window size based on receiver capability
- Uses physical layer clocking and synchronization to send and receive frames

HDLC Overview

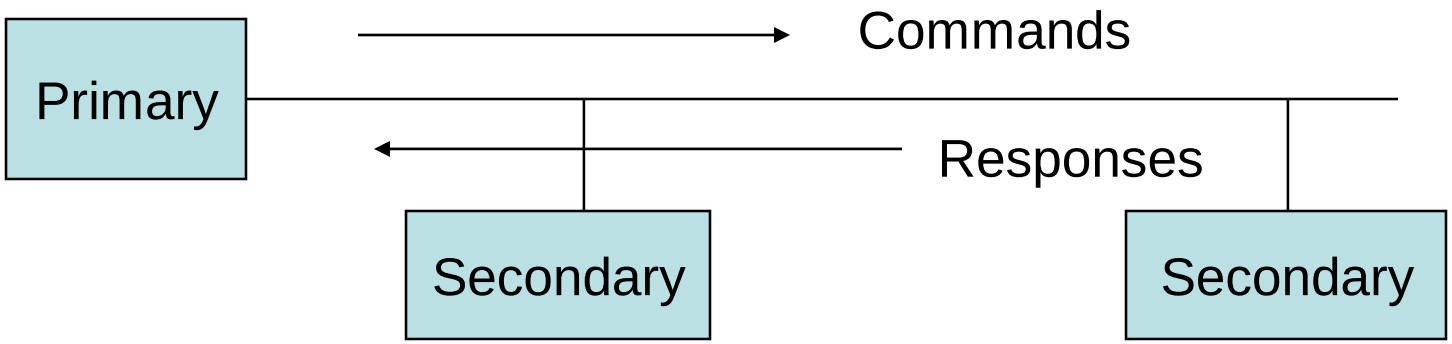
- Defines three types of stations
  - Primary
  - Secondary
  - Combined
- Defines three types of data transfer mode
  - Normal Response mode
  - Asynchronous Response mode
  - Asynchronous Balanced mode
- Three types of frames
  - Unnumbered
  - information
  - Supervisory

HDLC

- The three stations are :
  - Primary station
    - Has the responsibility of controlling the operation of data flow the link.
    - Handles error recovery
    - Frames issued by the primary station are called *commands.*
  - Secondary station,
    - Operates under the control of the primary station.
    - Frames issued by a secondary station are called *responses*.
    - The primary station maintains a separate logical link with each secondary station.
  - Combined station,
    - Acts as both as primary and secondary station.
    - Does not rely on other for sending data

HDLC

Unbalanced Mode



Balanced mode

HDLC

- The three modes of data transfer operations are
  - Normal Response Mode (NRM)
    - Mainly used in terminal-mainframe networks. In this case,
    - Secondaries (terminals) can only transmit when specifically instructed by the primary station in response to a polling
    - Unbalanced configuration, good for multi-point links
  - Asynchronous Response Mode (ARM)
    - Same as NRM except that the secondaries can initiate transmissions without direct polling from the primary station
    - Reduces overhead as no frames need to be sent to allow secondary nodes to transmit
    - Transmission proceeds when channel is detected idle , used mostly in point-to-point-links
  - Asynchronous Balanced Mode (ABM)
    - Mainly used in point-to-point links, for communication between combined stations

Non-operational Modes

- Normal Disconnected Mode
- Asynchronous Disconnected Mode

Both the above modes mean that the secondary node is logically disconnected from the primary node

- Initialization Mode
  - A node negotiates transmission parameters with the other node E.g., flow control information
  - Parameters negotiated in this mode are used during any of the data transfer modes

Data Link Control HDLC frame structure

Flag	Address	Control	Information	FCS	Flag
← 8 bits →	← 8 → Extendable	8 or 16 Extendable	← Variable →	16 or 32	← 8 →

(a) Frame format

(a) Frame Format

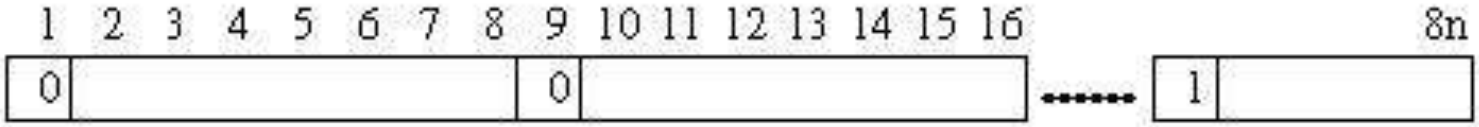
	1	2	3	4	5	6	7	8
I: Information	0	N(S)			P/F	N(R)		
S: Supervisory	1	0	S		P/F	N(R)		
U: Unnumbered	1	1	M		P/F	M		

N(S) = Send sequence number
N(R) = Receive sequence number
S = Supervisory function bits
M = Unnumbered function bits
P/F = Poll/final bit

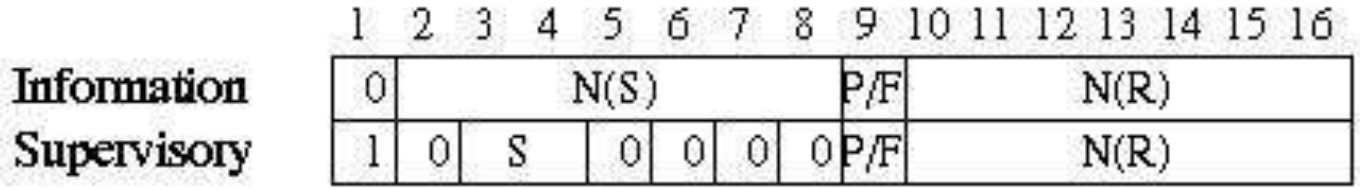
(b) Control field format

(b) Control field format

Data Link Control
HDLC frame structure



(c) Extended address field



(d) Extend control fields

(c) Extended address field (d) Extended control field

HDLC

- Flag: 01111110- start and ending delimiter. Bits are stuffed for flags in data frames
- FCS: 16-bit CRC using generating polynomial

$$G(x) = x^{16} + x^{12} + x^{5} + 1$$

- Address field:
  - mainly used in multidrop link configuration, and not used in point-to-point
  - In unbalanced configuration, every secondary is assigned a unique address. Contains address of secondary station in both command and response frames
  - In balanced mode, command frame has destination address and response frame has sending node's address
  - Group addresses are also possible. E.g., One command sent to all the secondaries
- In I-frames, N(s) is the sequence number of the frame being sent, and R(s) is the sequence number of the frame being expected.
- The P/F bit, known as the poll/final bit, is used with different meaning in different contexts.
  - It is used to indicate polling, to indicate the final I-frame, etc

HDLC

- There are three different classes of frames used in HDLC
  - Unnumbered frames, used in link setup and disconnection, and hence do not contain ACK.
  - Information frames, which carry actual information. Such frames can piggyback ACK in case of ABM
  - Supervisory frames, which are used for error and flow control purposes and hence contain send and receive sequence numbers

HDLC

- There are four different supervisory frames
  - SS=00, Receiver Ready (RR), and N(R) ACKs all frames received up to and including the one with sequence number N(R) - 1
  - SS=10, Receiver Not Ready (RNR), and N(R) has the same meaning as above
  - SS=01, Reject; all frames with sequence number N(R) or higher are rejected, which in turns ACKs frames with sequence number N(R) -1 or lower.
  - SS=11, Selective Reject; the receive rejects the frame with sequence number N(R)

HDLC

- The unnumbered frames can be grouped into the following categories:
  - Mode-setting commands and responses
  - Recovery commends and responses
  - Miscellaneous commands and responses

Review of Link Layer

- Services
  - Framing
  - Error control
  - Reliability
  - Connection management
  - Medium access control
  - Switching
- Protocols, Standards
  - Ethernet
  - Token Ring
  - FDDI
  - Wireless
  - PPP
  - HDLC